



A UGC-QAA Certified Institution

NIET

National Institute of Engineering and Technology

Affiliated to Purbanchal University

New Beginning ...

Endless Possibilities !!!



BE BIOMEDICAL

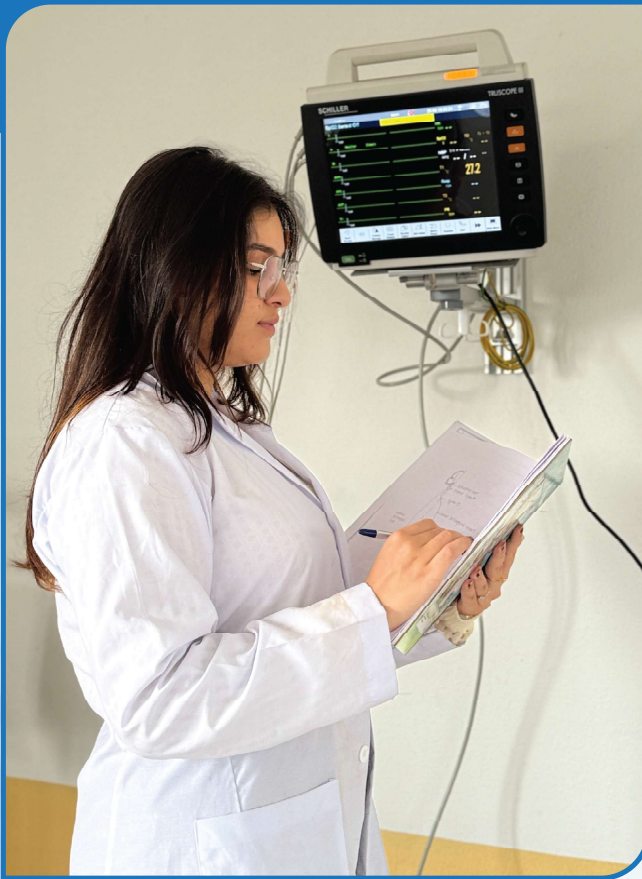
Bachelor of Engineering in Biomedical

Offered Programs:

BE Biomedical

BE Computer

B. Tech in AI



INTRODUCTION

National Institute of Engineering and Technology (NIET), former College of Biomedical Engineering and Applied Sciences (CBEAS), established in 2005, is affiliated to Purbanchal University and registered with Nepal Engineering Council. There are only a few colleges in Asia providing higher education in Biomedical Engineering and this is the only one in Nepal. Within a very short span of time, this college has established a coveted place in this field and efforts continue to make it a unique institution in the field of biomedical education. The college uses all its initiatives in facilitating new opportunities for students and gain a new height.

The college is imparting world class education providing professional opportunities here and abroad.





BE Biomedical

Bachelor in Biomedical Engineering

Biomedical Engineering is the application of engineering principles to biology and medicine, combining the design and problem-solving techniques of engineering with medical sciences for advanced healthcare treatment. It covers a wide range of areas from cell and tissue engineering to modeling of human organs, implantable devices, diagnostics, and therapeutic medical devices and equipment.

Why Biomedical Engineering?

- ★ Prepare & Craft the MedTech Problem Solver Engineer
- ★ Analyze Novel Solutions to Complex Medical Problems
- ★ Develop Real-time Medical Technology
- ★ Focus on improving the Quality & Length of Human Life
- ★ Build & Maintain the Complex Array of Medical Equipment
- ★ Become an Interdisciplinary Communicator for Engineering & Medical Science



Course Structure

Semester I	
S.N	Subject
1	Mathematics I
2	Physics
3	Basic Electronics Engineering
4	Computer Programming
5	Engineering Drawing
6	Basic Electrical Engineering

Semester II	
S.N	Subject
1	Mathematics II
2	Chemistry
3	Basic Mechanical Engineering
4	Electro-engineering Materials
5	Introduction to Computational System
6	Digital Electronics

Semester III	
S.N	Subject
1	Mathematics III
2	Bio-Engineering Materials & Components
3	Human Anatomy and Physiology I
4	Fluid Mechanics & Thermodynamics
5	Cell Biology and Immunology
6	Microprocessors

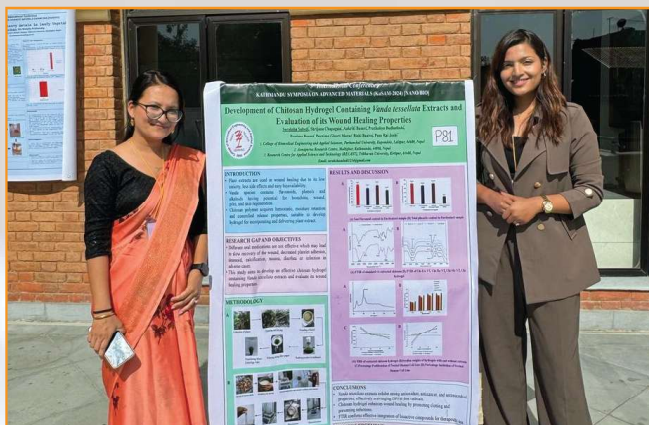
Semester IV	
S.N	Subject
1	Applied Mathematics
2	Electronic Devices and Circuits
3	Applied Sociology
4	Biomechanics
5	Biomedical Embedded System Design
6	Human Anatomy and Physiology II

Semester V	
S.N	Subject
1	Probability and Statistics
2	Numerical Methods
3	Implantable Devices
4	Control Systems
5	Tissue Device Interactions
6	Communication Systems

Semester VI	
S.N	Subject
1	Engineering Economics
2	Medical Industry Management
3	Medical Imaging I
4	Biomedical Instrumentation I
5	Medical Electronics
6	Biomedical Digital Signal Processing

Semester VII	
S.N	Subject
1	Organization & Project Management
2	Biomedical Instrumentation II
3	Medical Imaging II
4	Elective I : Structural & Functional Biomaterials, Biomedical Equipment Maintenance I, Physiological Modeling
5	Elective II : Minimally Invasive Medical Technology, Medical Image Processing, Principles of Tissue Engineering, Theory of Medical Robotics
6	Project

Semester VIII	
S.N	Subject
1	Engineering Professional Practice
2	Elective III : Neural Network, Medical Informatics, Telemedicine and Telehealth, Biomedical Applications of Nanotechnology, Biomedical Equipment Maintenance II, Bio-Electromagnetism
3	Internship
4	Project

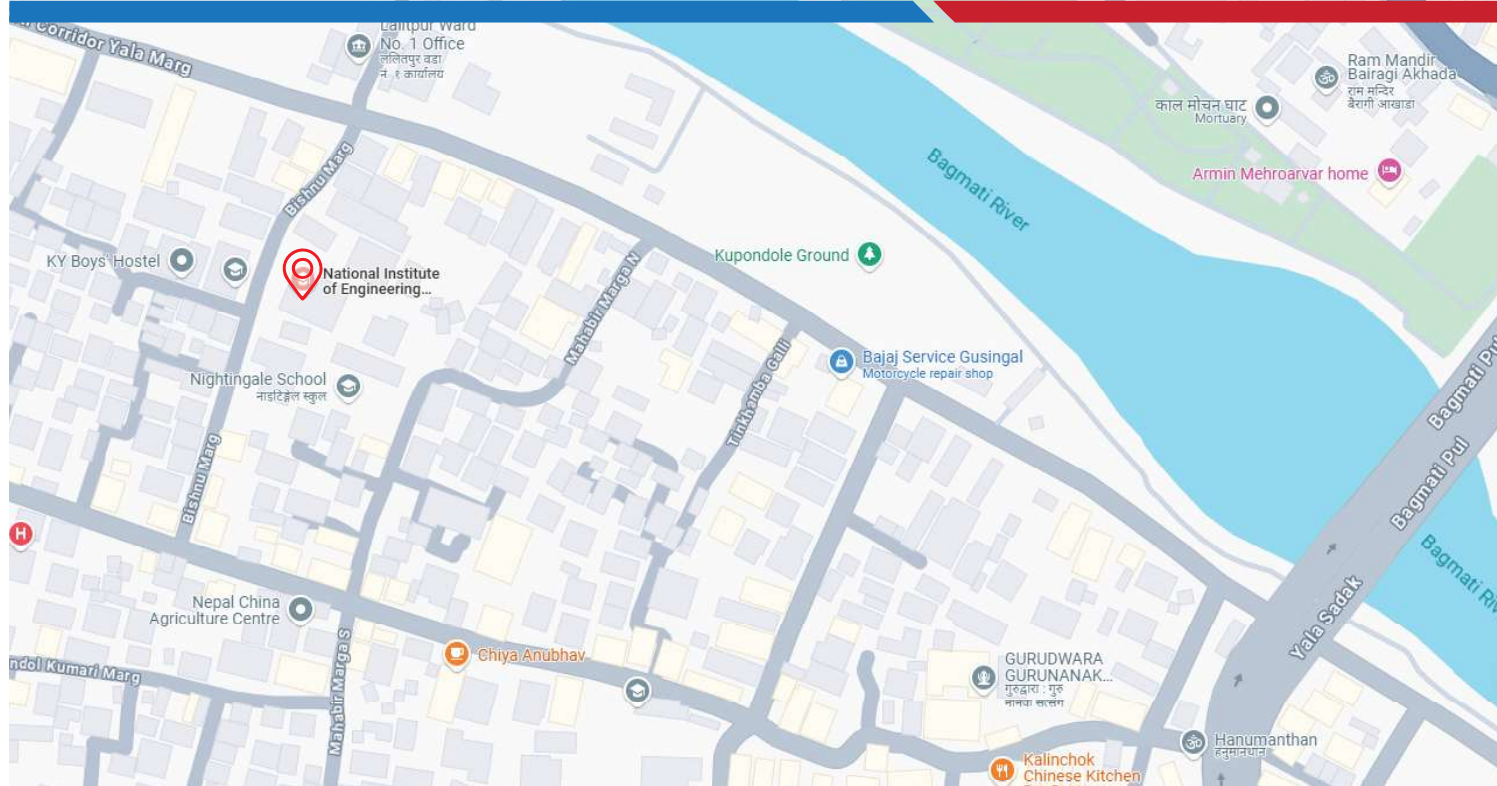


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Find us at
Kupondole, Lalitpur
(Bagmati Corridor)



Call us for more
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